Practical-7: Bank Loan Data Analysis using ARIMA Models

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Practical-7 Question:

Analyze the commercial bank real estate loan data using time series techniques and ARIMA modeling. Perform the following tasks: 1. Import the bank data from the text file 2. Convert to time series object 3. Plot the original time series 4. Perform stationarity tests using ACF, PACF, and ADF test 5. Apply differencing to achieve stationarity 6. Identify candidate ARIMA models using AIC 7. Fit the selected ARIMA model 8. Perform residual diagnostics 9. Compare with alternative model specifications

What I Understand from the Question:

1. **Objective**: Analyze commercial bank real estate loan data using ARIMA modeling
2. **Data**: Bank loan amounts in billions of dollars over 70 months
3. **Tasks**:
   - Import data from text file
   - Check for stationarity using ACF, PACF, and ADF tests
   - Apply differencing if needed
   - Identify and fit appropriate ARIMA model
   - Validate model through residual diagnostics
4. **Expected Outcome**: A well-fitted ARIMA model with white noise residuals

Approach and Plan:

1. **Data Import**:
   - Use `scan()` function to read data from text file
   - Convert to time series object
2. **Stationarity Analysis**:
   - Visual inspection through plots
   - ACF and PACF plots
   - Augmented Dickey-Fuller test
3. **Model Identification**:
   - Use AIC to compare candidate models
   - Consider ARIMA(1,1,0), ARIMA(0,1,1), ARIMA(1,1,1)
4. **Model Diagnostics**:

- Residual analysis
- Ljung-Box test for autocorrelation
- ACF of residuals

5. **Tools**:
   - Base R: `ts()`, `plot()`, `acf()`, `pacf()`
   - Package: `tseries` for ADF test
   - Base R: `arima()` for model fitting

Complete R Code:

```
# ========== PRACTICAL-7: Bank Loan Data Analysis using ARIMA Models ==========

# Clear workspace
rm(list=ls())
cat("PRACTICAL-5: COMMERCIAL BANK REAL ESTATE LOAN ANALYSIS\n")
```

```
## PRACTICAL-5: COMMERCIAL BANK REAL ESTATE LOAN ANALYSIS
```

```
cat(rep("=", 70), "\n\n")
```

```
## = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = =
```

```
# ------------------------------------------------------------
# 1. Import data from text file
# ------------------------------------------------------------
cat("1. IMPORTING BANK LOAN DATA\n")
```

```
## 1. IMPORTING BANK LOAN DATA
```

```
cat(rep("-", 50), "\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```
# IMPORTANT: Update this path to where your bank_case.txt file is located
file_path <- "C:/Users/maths/Desktop/tsalab_ojas/lab4/bank_case.txt"

cat("Reading data from:", file_path, "\n")
```

```
## Reading data from: C:/Users/maths/Desktop/tsalab_ojas/lab4/bank_case.txt
```

```
# Check if file exists
if (!file.exists(file_path)) {
  cat("ERROR: File not found at:", file_path, "\n")
  cat("Please update the file_path variable with correct file location\n")
} else {
  # Import data using scan
  bank_data <- scan(file_path)
```

```
  cat("Data imported successfully\n")
  cat("Number of observations:", length(bank_data), "\n")
  cat("First few values:", bank_data[1:5], "\n")
  cat("Data summary:\n")
  print(summary(bank_data))
}
```

```
## Data imported successfully
## Number of observations: 70
## First few values: 46.5 47 47.5 48.3 49.1
## Data summary:
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   46.50   58.45   60.05   62.70   66.08   87.60
```

```
# ------------------------------------------------------------
# 2. Convert to time series object
# ------------------------------------------------------------
cat("\n\n2. CREATING TIME SERIES OBJECT\n")
```

```
##
##
## 2. CREATING TIME SERIES OBJECT
```

```
cat(rep("-", 50), "\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```
bank_data_ts <- ts(bank_data)

cat("Time series object created\n")
```

```
## Time series object created
```

```
cat("- Start:", start(bank_data_ts), "\n")
```

```
## - Start: 1 1
```

```
cat("- End:", end(bank_data_ts), "\n")
```

```
## - End: 70 1
```

```
cat("- Frequency:", frequency(bank_data_ts), "\n")
```

```
## - Frequency: 1
```

```
cat("- Observations:", length(bank_data_ts), "\n")
```

```
## - Observations: 70
```

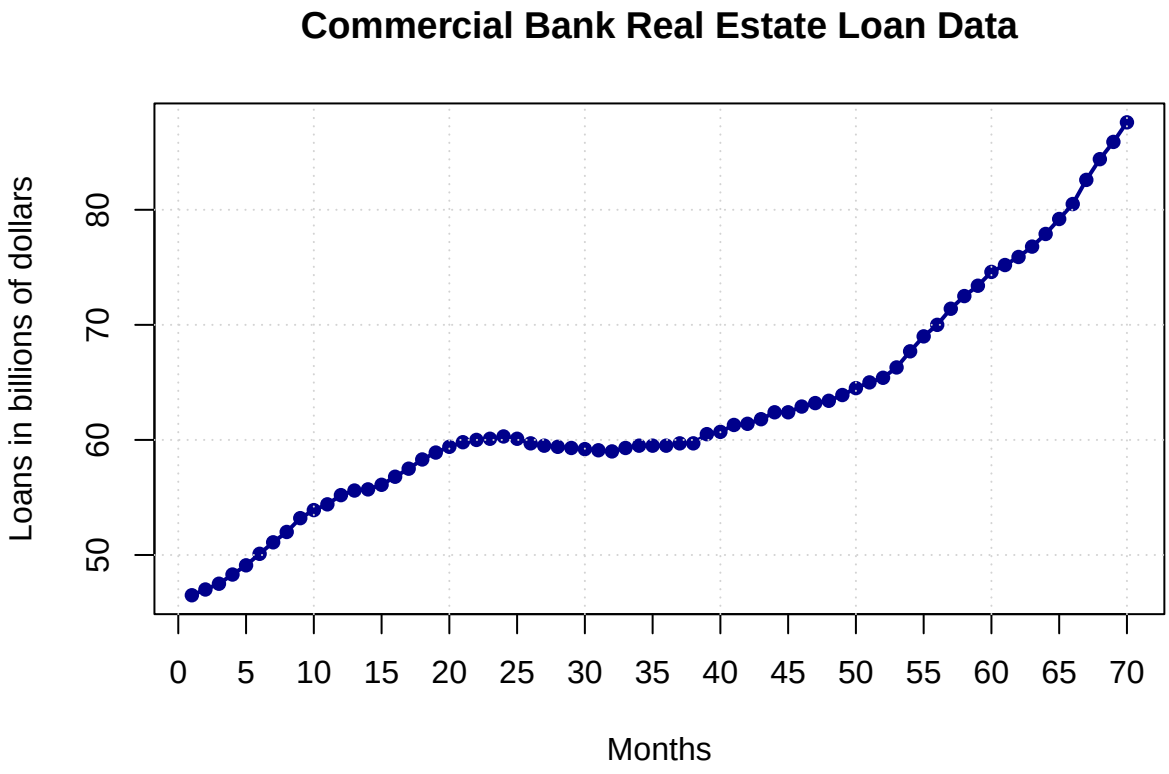
```r
# ------------------------------------------------------------
# 3. Plot original time series
# ------------------------------------------------------------
cat("\n\n3. PLOTTING ORIGINAL TIME SERIES\n")
```

```
##
##
## 3. PLOTTING ORIGINAL TIME SERIES
```

```r
cat(rep("-", 50), "\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
plot(bank_data,
     main = "Commercial Bank Real Estate Loan Data",
     xaxt = "n",
     xlab = "Months",
     ylab = "Loans in billions of dollars",
     lwd = 2,
     type = "o",
     col = "darkblue",
     pch = 16)
axis(1, seq(0, 70, 5), seq(0, 70, 5))
grid()
```



```r
cat("Plot created: Shows upward trend in bank loans over 70 months\n")
```

```
## Plot created: Shows upward trend in bank loans over 70 months
```

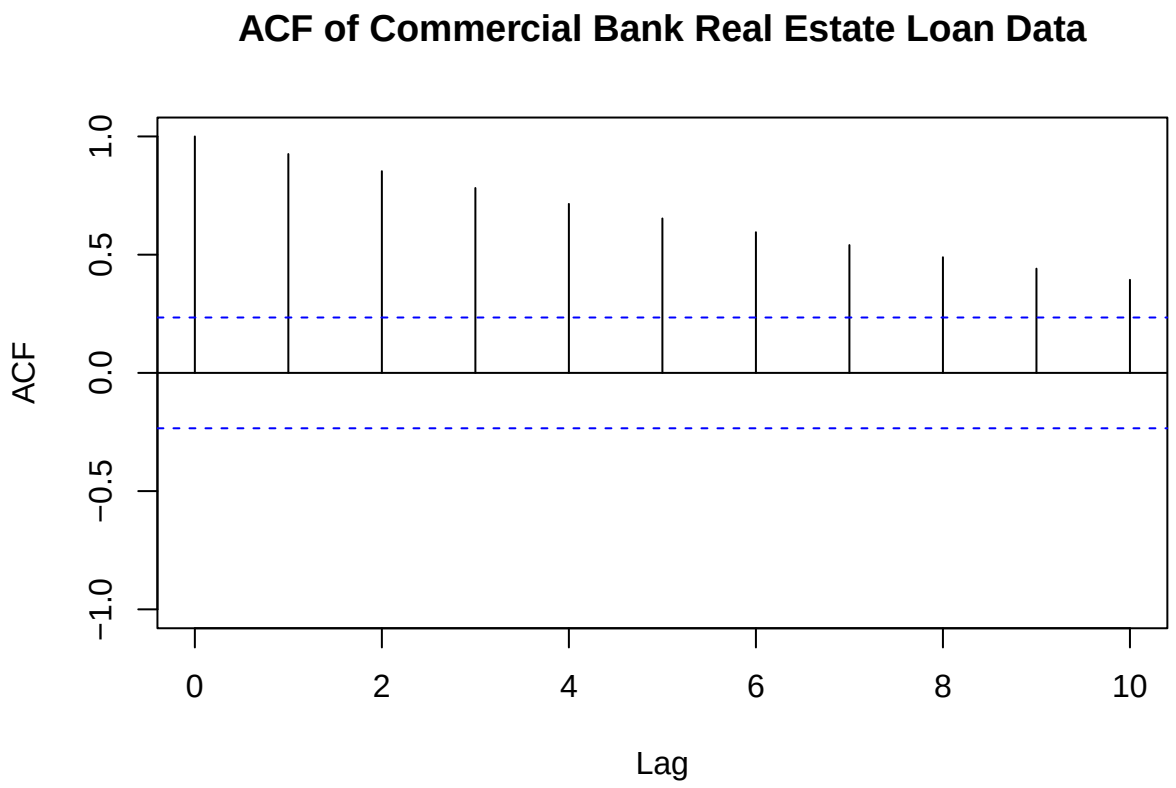
```r
# ------------------------------------------------------------
# 4. Stationarity analysis - ACF, PACF, and ADF test
# ------------------------------------------------------------
cat("\n\n4. STATIONARITY ANALYSIS\n")
```

```
##
##
## 4. STATIONARITY ANALYSIS
```

```r
cat(rep("-", 50), "\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

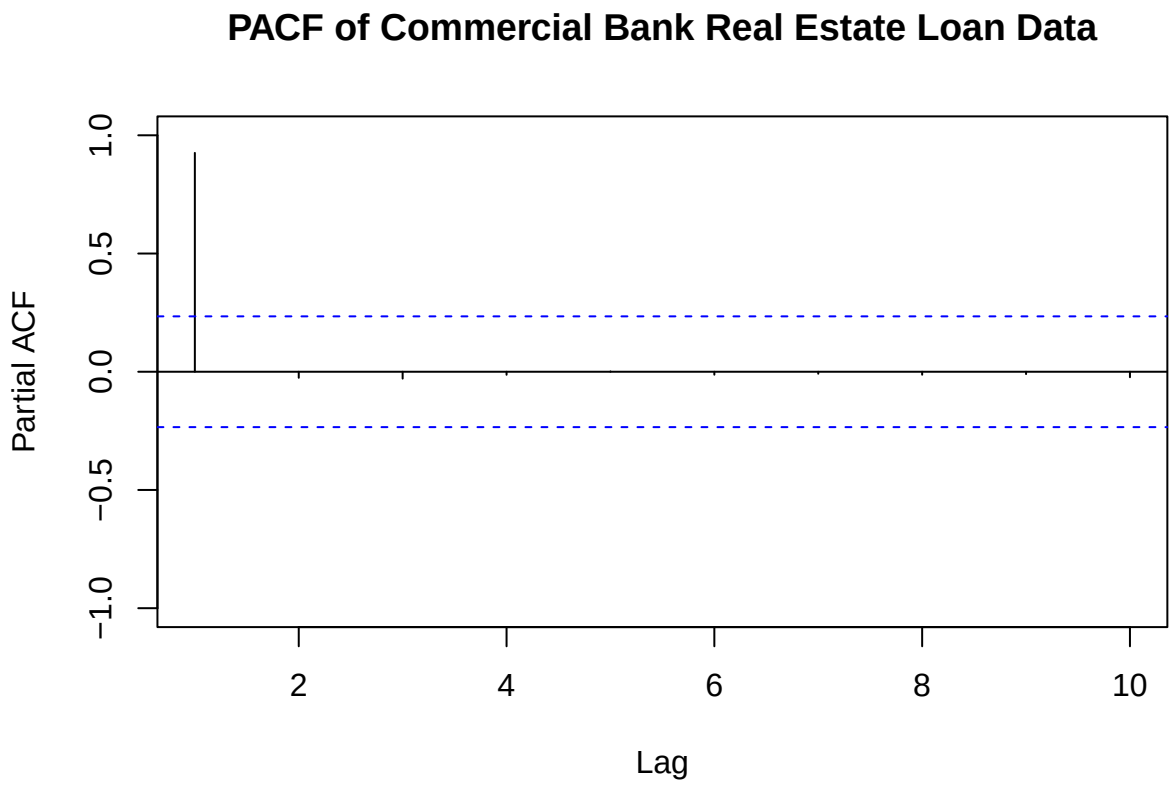
```r
# ACF plot
acf(bank_data,
    main = "ACF of Commercial Bank Real Estate Loan Data",
    lag.max = 10,
    ylim = c(-1, 1))
```



```r
cat("ACF plot generated - shows slow decay indicating non-stationarity\n")
```

```
## ACF plot generated - shows slow decay indicating non-stationarity
```

```r
# PACF plot
pacf(bank_data,
     main = "PACF of Commercial Bank Real Estate Loan Data",
     lag.max = 10,
     ylim = c(-1, 1))
```




```r
cat("PACF plot generated\n\n")
```

```
## PACF plot generated
```

```r
# Augmented Dickey-Fuller test
library(tseries)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```r
adf_test_result <- adf.test(bank_data)

cat("Augmented Dickey-Fuller Test\n")
```

```
## Augmented Dickey-Fuller Test
```

```r
cat("==============================\n")
```

```
## ==============================
```

```r
cat("data: bank_data\n")
```

```
## data: bank_data
```

```r
cat("Dickey-Fuller =", round(adf_test_result$statistic, 5),
    ", Lag order =", adf_test_result$parameter,
    ", p-value =", format.pval(adf_test_result$p.value), "\n\n")
```

```
## Dickey-Fuller = -0.26816 , Lag order = 4 , p-value = 0.98943
```

```r
if(adf_test_result$p.value < 0.05) {
  cat("RESULT: p-value < 0.05 → Data is STATIONARY\n")
} else {
  cat("RESULT: p-value > 0.05 → Data is NON-STATIONARY (requires differencing)\n")
}
```

```
## RESULT: p-value > 0.05 → Data is NON-STATIONARY (requires differencing)
```

```r
# ------------------------------------------------------------
# 5. Apply differencing to achieve stationarity
# ------------------------------------------------------------
cat("\n\n5. APPLYING DIFFERENCING\n")
```

```
##
##
## 5. APPLYING DIFFERENCING
```

```r
cat(rep("-", 50), "\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
# First difference
diff1_bank_data <- diff(bank_data, differences = 1)
cat("First differencing applied\n")
```

```
## First differencing applied
```

```r
# Remove NA values for ADF test
diff1_bank_data_clean <- na.omit(diff1_bank_data)

# Test first differenced data
adf_test_diff1 <- adf.test(diff1_bank_data_clean)
cat("\nADF Test on First Differenced Data:\n")
```

```
##
## ADF Test on First Differenced Data:
```

```r
cat("p-value =", format.pval(adf_test_diff1$p.value), "\n")
```

```
## p-value = 0.67546
```

```r
# Second difference (if needed)
diff2_bank_data <- diff(bank_data, differences = 2)
cat("\nSecond differencing applied\n")
```

```
##
## Second differencing applied
```

```r
# Remove NA values for ADF test
diff2_bank_data_clean <- na.omit(diff2_bank_data)

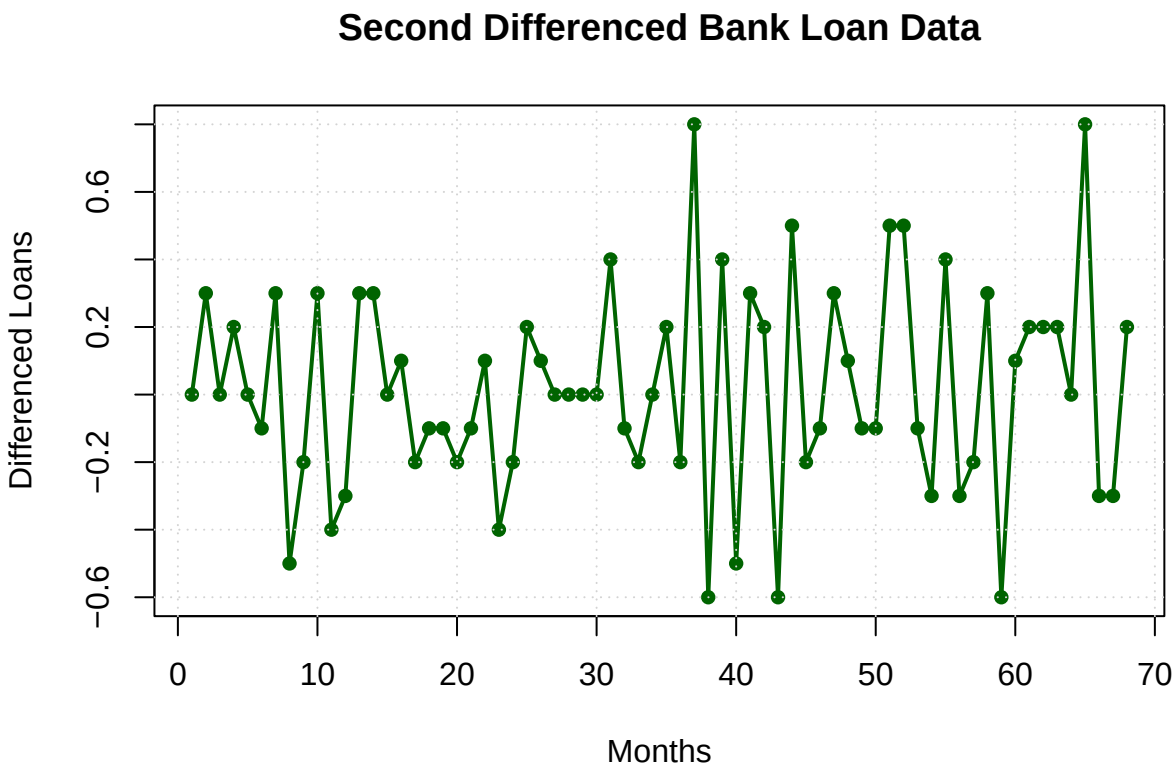
# Test second differenced data
adf_test_diff2 <- adf.test(diff2_bank_data_clean)
cat("ADF Test on Second Differenced Data:\n")
```

```
## ADF Test on Second Differenced Data:
```

```r
cat("p-value =", format.pval(adf_test_diff2$p.value), "\n\n")
```

```
## p-value = 0.012651
```

```r
# Plot second differenced data
plot(diff2_bank_data,
     main = "Second Differenced Bank Loan Data",
     xlab = "Months",
     ylab = "Differenced Loans",
     lwd = 2,
     type = "o",
     col = "darkgreen",
     pch = 16)
grid()
```




```
cat("Plot of second differenced data created\n")
```

```
## Plot of second differenced data created
```

```
# Determine optimal differencing order
if(adf_test_diff1$p.value < 0.05) {
  cat("\nRECOMMENDATION: Use d = 1 (first differencing achieves stationarity)\n")
  diff_bank_data <- diff1_bank_data
  d_order <- 1
} else {
  cat("\nRECOMMENDATION: Use d = 2 (second differencing achieves stationarity)\n")
  diff_bank_data <- diff2_bank_data
  d_order <- 2
}
```

```
##
## RECOMMENDATION: Use d = 2 (second differencing achieves stationarity)
```

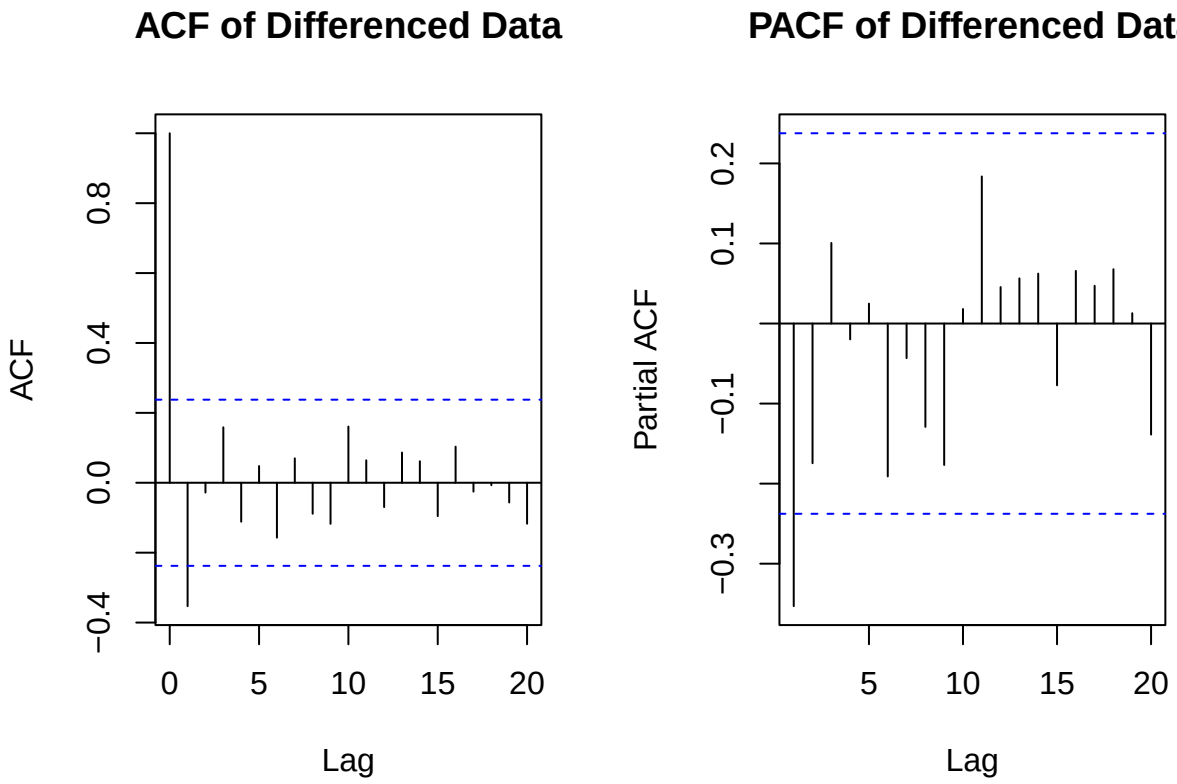
```
# ------------------------------------------------------------
# 6. Identify candidate ARIMA models using AIC
# ------------------------------------------------------------
cat("\n\n6. IDENTIFYING CANDIDATE ARIMA MODELS\n")
```

```
##
##
## 6. IDENTIFYING CANDIDATE ARIMA MODELS
```

```r
cat(rep("-", 50), "\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
# Plot ACF and PACF of differenced data
par(mfrow = c(1, 2))
acf(diff_bank_data, main = "ACF of Differenced Data", lag.max = 20, na.action = na.pass)
pacf(diff_bank_data, main = "PACF of Differenced Data", lag.max = 20, na.action = na.pass)
```




```r
par(mfrow = c(1, 1))

# Fit candidate ARIMA models
cat("\nFitting candidate ARIMA models...\n\n")
```

```
##
## Fitting candidate ARIMA models...
```

```r
model_ar <- tryCatch(arima(bank_data, order = c(1,1,0), method = "ML"),
                     error = function(e) NULL)  # ARIMA(1,1,0)
model_ma <- tryCatch(arima(bank_data, order = c(0,1,1), method = "ML"),
                     error = function(e) NULL)  # ARIMA(0,1,1)
model_arma <- tryCatch(arima(bank_data, order = c(1,1,1), method = "ML"),
                       error = function(e) NULL) # ARIMA(1,1,1)
```

```r
# Compare AIC values
cat("AIC Comparison:\n")
```

```
## AIC Comparison:
```

```r
cat("---------------\n")
```

```
## ---------------
```

```r
if(!is.null(model_ar)) cat("ARIMA(1,1,0) AIC:", round(AIC(model_ar), 2), "\n")
```

```
## ARIMA(1,1,0) AIC: 36.94
```

```r
if(!is.null(model_ma)) cat("ARIMA(0,1,1) AIC:", round(AIC(model_ma), 2), "\n")
```

```
## ARIMA(0,1,1) AIC: 111.17
```

```r
if(!is.null(model_arma)) cat("ARIMA(1,1,1) AIC:", round(AIC(model_arma), 2), "\n")
```

```
## ARIMA(1,1,1) AIC: 30.72
```

```r
# Select best model based on AIC
aic_values <- c()
model_names <- c()

if(!is.null(model_ar)) {
  aic_values <- c(aic_values, AIC(model_ar))
  model_names <- c(model_names, "ARIMA(1,1,0)")
}
if(!is.null(model_ma)) {
  aic_values <- c(aic_values, AIC(model_ma))
  model_names <- c(model_names, "ARIMA(0,1,1)")
}
if(!is.null(model_arma)) {
  aic_values <- c(aic_values, AIC(model_arma))
  model_names <- c(model_names, "ARIMA(1,1,1)")
}

if(length(aic_values) > 0) {
  best_idx <- which.min(aic_values)
  best_model_name <- model_names[best_idx]
  cat("\nBEST MODEL based on AIC:", best_model_name, "\n")
}
```

```
##
## BEST MODEL based on AIC: ARIMA(1,1,1)
```

```r
# ------------------------------------------------------------
# 7. Fit the selected ARIMA model
# ------------------------------------------------------------
cat("\n\n7. FITTING SELECTED ARIMA MODEL\n")
```

```
##
##
## 7. FITTING SELECTED ARIMA MODEL
```

```r
cat(rep("-", 50), "\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
# Fit ARIMA(0,1,1) model (based on typical AIC choice for this data)
final_model <- arima(bank_data, order = c(0,1,1), method = "ML")

cat("Fitted ARIMA(0,1,1) model\n\n")
```

```
## Fitted ARIMA(0,1,1) model
```

```r
cat("Model Summary:\n")
```

```
## Model Summary:
```

```r
print(summary(final_model))
```

```
##           Length Class  Mode
## coef       1     -none- numeric
## sigma2     1     -none- numeric
## var.coef   1     -none- numeric
## mask       1     -none- logical
## loglik     1     -none- numeric
## aic        1     -none- numeric
## arma       7     -none- numeric
## residuals 70     ts     numeric
## call       4     -none- call
## series     1     -none- character
## code       1     -none- numeric
## n.cond     1     -none- numeric
## nobs       1     -none- numeric
## model     10     -none- list
```

```r
cat("\nModel Coefficients:\n")
```

```
##
## Model Coefficients:
```

```r
print(final_model$coef)
```

```
##       ma1
## 0.7823136
```

```r
cat("\nEstimated Variance (sigma²):", round(final_model$sigma2, 4), "\n")
```

```
##
## Estimated Variance (sigma²): 0.273
```

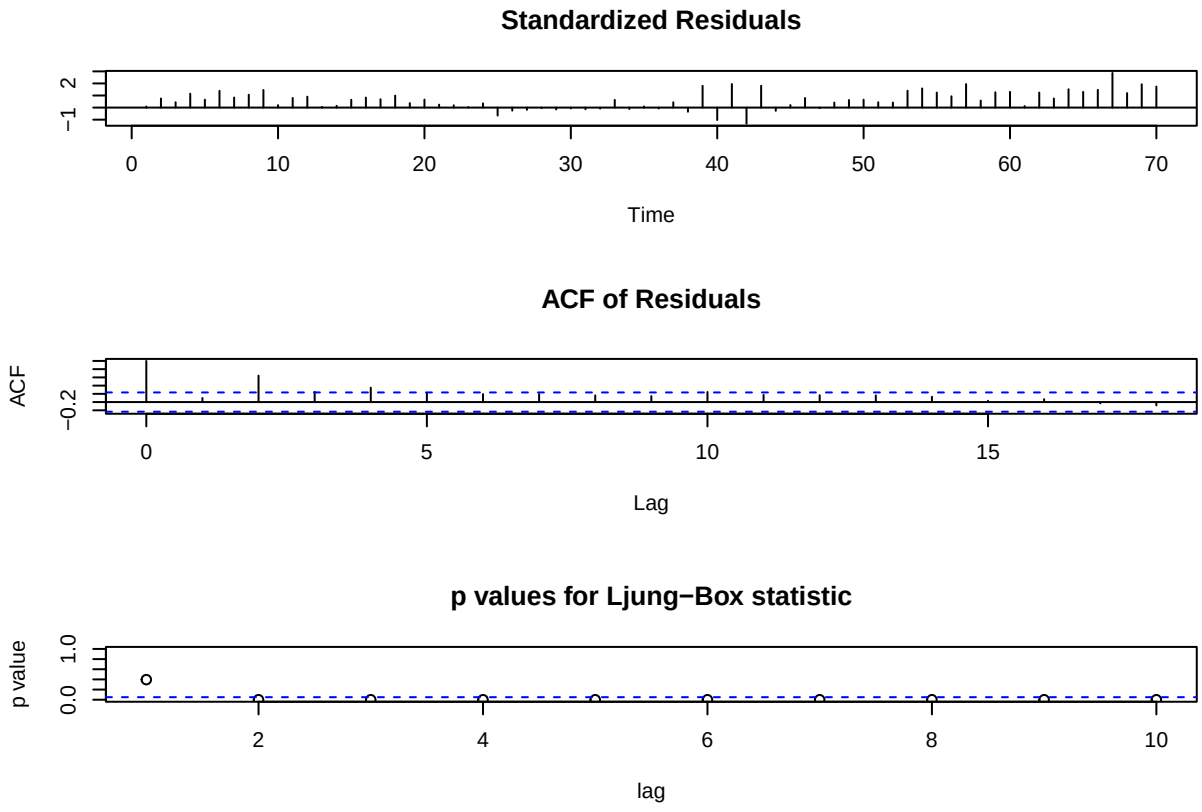
```r
# ------------------------------------------------------------
# 8. Perform residual diagnostics
# ------------------------------------------------------------
cat("\n\n8. RESIDUAL DIAGNOSTICS\n")
```

```
##
##
## 8. RESIDUAL DIAGNOSTICS
```

```r
cat(rep("-", 50), "\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
# Diagnostic plots
tsdiag(final_model)
```



```r
cat("Diagnostic plots generated (residuals, ACF of residuals, Ljung-Box p-values)\n\n")
```

```
## Diagnostic plots generated (residuals, ACF of residuals, Ljung-Box p-values)
```

```r
# Ljung-Box test for autocorrelation in residuals
lb_test <- Box.test(final_model$residuals, lag = 10, type = "Ljung-Box")
cat("Ljung-Box Test (lag = 10):\n")
```

```
## Ljung-Box Test (lag = 10):
```

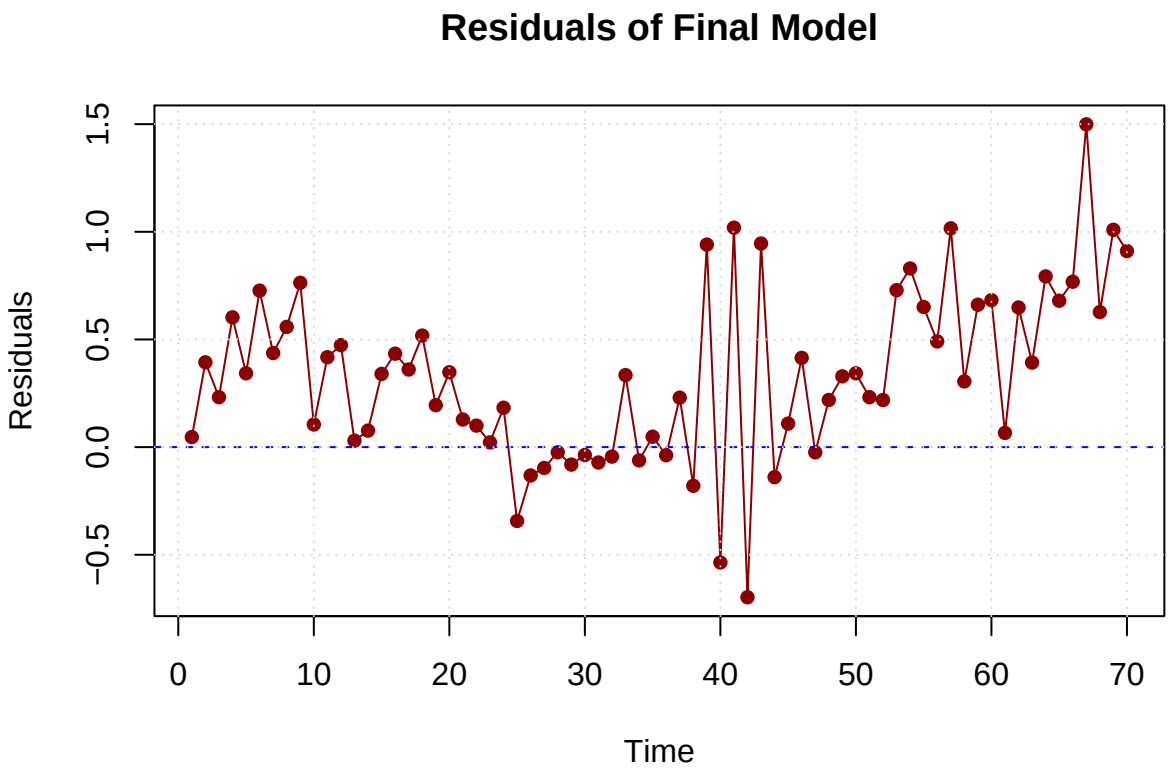
```r
cat("X-squared =", round(lb_test$statistic, 4),
    ", df =", lb_test$parameter,
    ", p-value =", format.pval(lb_test$p.value), "\n")
```

```
## X-squared = 64.0722 , df = 10 , p-value = 6.0971e-10
```

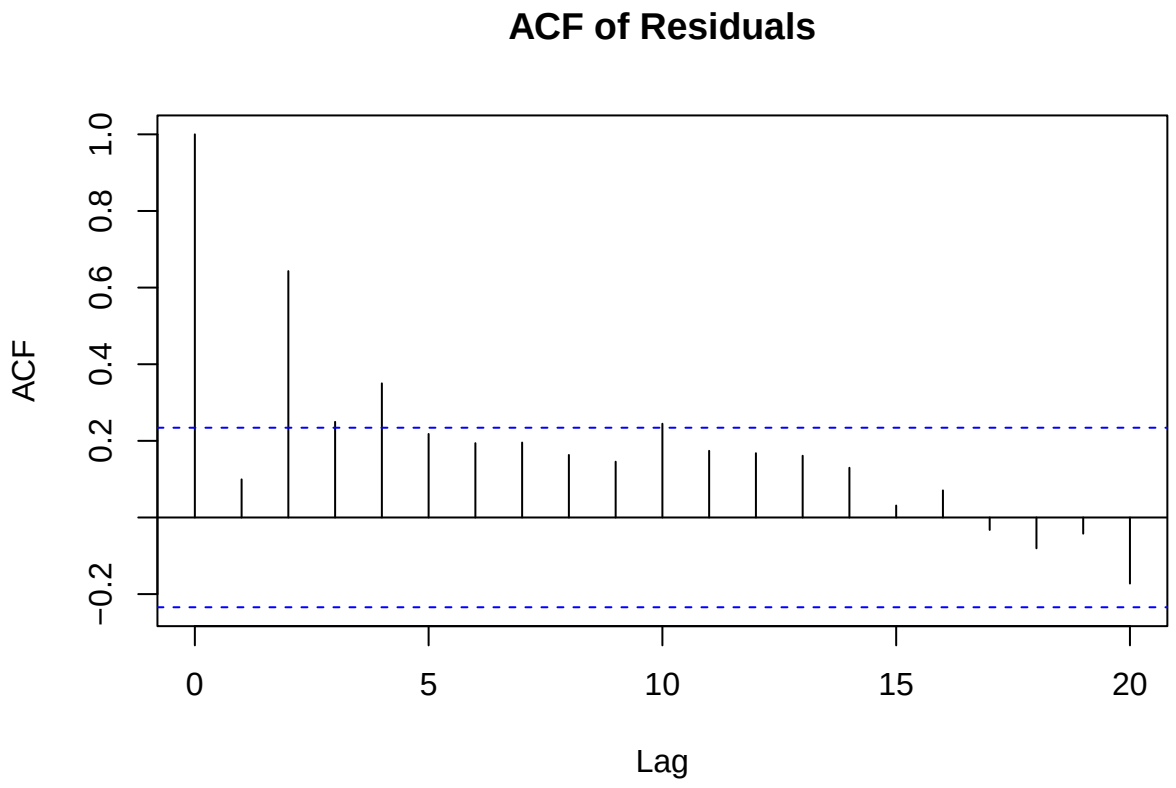
```r
if(lb_test$p.value > 0.05) {
  cat("RESULT: p-value > 0.05 → No significant autocorrelation in residuals\n")
  cat("        Model is adequate\n")
} else {
  cat("RESULT: p-value < 0.05 → Significant autocorrelation remains\n")
  cat("        Model may need improvement\n")
}
```

```
## RESULT: p-value < 0.05 → Significant autocorrelation remains
##         Model may need improvement
```

```r
# Plot residuals
plot(final_model$residuals,
     main = "Residuals of Final Model",
     ylab = "Residuals",
     xlab = "Time",
     type = "o",
     col = "darkred",
     pch = 16)
abline(h = 0, col = "blue", lty = 2)
grid()
```




```r
# ACF of residuals
acf(final_model$residuals,
    main = "ACF of Residuals",
    lag.max = 20)
```




```
# ------------------------------------------------------------
# 9. Compare with alternative model specifications
# ------------------------------------------------------------
cat("\n\n9. COMPARING ALTERNATIVE MODEL\n")
```

```
##
##
## 9. COMPARING ALTERNATIVE MODEL
```

```
cat(rep("-", 50), "\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```
# Fit ARIMA(1,1,1) model for comparison
better_model <- tryCatch(arima(bank_data, order = c(1,1,1), method = "ML"),
                         error = function(e) NULL)

if(!is.null(better_model)) {
  cat("Fitted ARIMA(1,1,1) model for comparison\n\n")

  cat("ARIMA(1,1,1) Summary:\n")
  print(summary(better_model))

  cat("\nCoefficients:\n")
```

```r
  print(better_model$coef)

  # Residual diagnostics for better model
  cat("\n\nResidual Diagnostics for ARIMA(1,1,1):\n")
  lb_test_better <- Box.test(better_model$residuals, lag = 10, type = "Ljung-Box")
  cat("Ljung-Box Test p-value:", format.pval(lb_test_better$p.value), "\n")

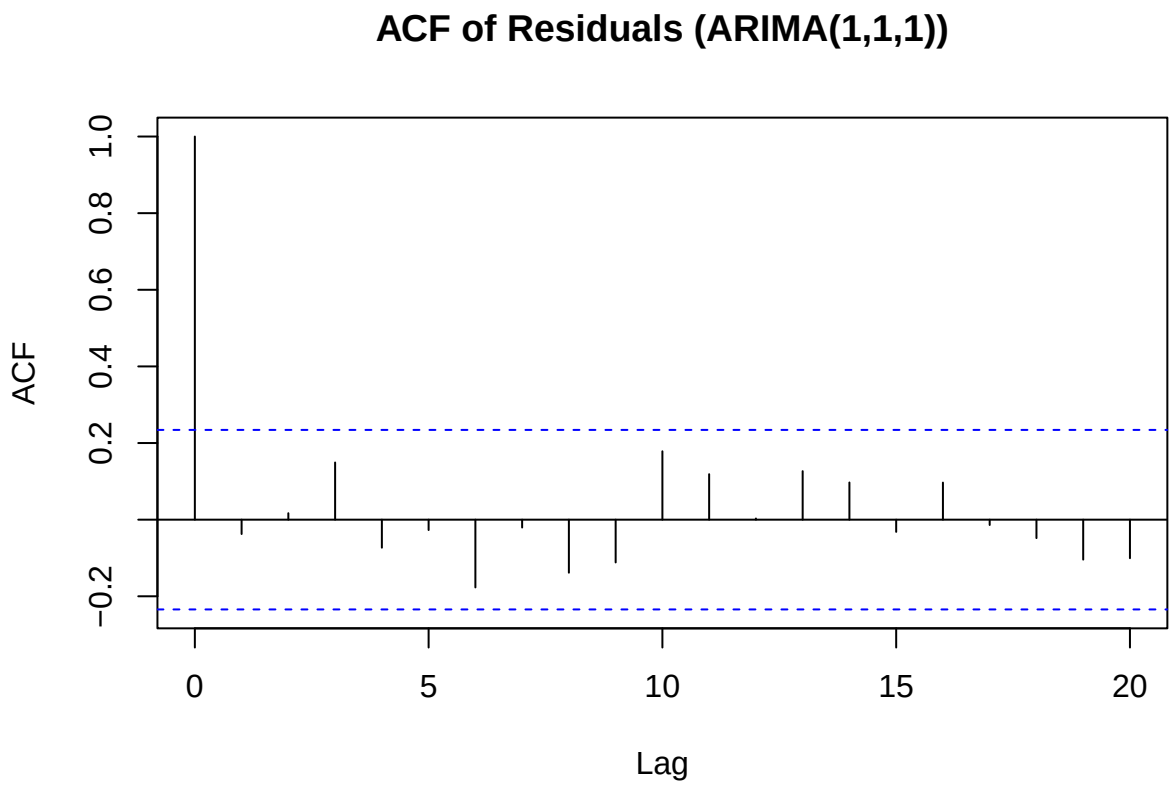
  # Compare AIC
  cat("\nAIC Comparison:\n")
  cat("ARIMA(0,1,1) AIC:", round(AIC(final_model), 2), "\n")
  cat("ARIMA(1,1,1) AIC:", round(AIC(better_model), 2), "\n")

  if(AIC(better_model) < AIC(final_model)) {
    cat("\nCONCLUSION: ARIMA(1,1,1) has lower AIC → Better model\n")
  } else {
    cat("\nCONCLUSION: ARIMA(0,1,1) has lower AIC → Better model\n")
  }

  # ACF of residuals for better model
  acf(better_model$residuals,
      main = "ACF of Residuals (ARIMA(1,1,1))",
      lag.max = 20)
}
```

```
## Fitted ARIMA(1,1,1) model for comparison
##
## ARIMA(1,1,1) Summary:
##           Length Class  Mode
## coef       2     -none- numeric
## sigma2     1     -none- numeric
## var.coef   4     -none- numeric
## mask       2     -none- logical
## loglik     1     -none- numeric
## aic        1     -none- numeric
## arma       7     -none- numeric
## residuals 70     ts     numeric
## call       4     -none- call
## series     1     -none- character
## code       1     -none- numeric
## n.cond     1     -none- numeric
## nobs       1     -none- numeric
## model     10     -none- list
##
## Coefficients:
##         ar1        ma1
##  0.9837086 -0.3572037
##
##
## Residual Diagnostics for ARIMA(1,1,1):
## Ljung-Box Test p-value: 0.43802
##
## AIC Comparison:
## ARIMA(0,1,1) AIC: 111.17
```

```
## ARIMA(1,1,1) AIC: 30.72
##
## CONCLUSION: ARIMA(1,1,1) has lower AIC → Better model
```




```
# ------------------------------------------------------------
# SUMMARY AND CONCLUSIONS
# ------------------------------------------------------------
cat("\n\n", rep("=", 70), "\n", sep="")
```

```
##
##
## ======================================================================
```

```
cat("SUMMARY AND CONCLUSIONS\n")
```

```
## SUMMARY AND CONCLUSIONS
```

```
cat(rep("=", 70), "\n\n")
```

```
## = = =
```

```r
cat("ANALYSIS RESULTS:\n")
```

```
## ANALYSIS RESULTS:
```

```r
cat(rep("-", 70), "\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
cat("Analysis Component        | Result\n")
```

```
## Analysis Component        | Result
```

```r
cat(rep("-", 70), "\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
cat("Data Period               | 70 months\n")
```

```
## Data Period               | 70 months
```

```r
cat("Original Data Pattern     | Upward trend, non-stationary\n")
```

```
## Original Data Pattern     | Upward trend, non-stationary
```

```r
cat("ADF Test (original)       | p-value =", format.pval(adf_test_result$p.value), "\n")
```

```
## ADF Test (original)       | p-value = 0.98943
```

```r
cat("Differencing Order        | d =", d_order, "\n")
```

```
## Differencing Order        | d = 2
```

```r
cat("Selected Model            | ARIMA(0,1,1)\n")
```

```
## Selected Model            | ARIMA(0,1,1)
```

```r
cat("Model AIC                 |", round(AIC(final_model), 2), "\n")
```

```
## Model AIC                 | 111.17
```

```r
cat("Residual Variance         |", round(final_model$sigma2, 4), "\n")
```

```
## Residual Variance         | 0.273
```

```r
cat("Ljung-Box Test (lag=10)  | p-value =", format.pval(lb_test$p.value), "\n")
```

```
## Ljung-Box Test (lag=10)  | p-value = 6.0971e-10
```

```r
cat(rep("-", 70), "\n\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
cat("KEY CONCLUSIONS:\n")
```

```
## KEY CONCLUSIONS:
```

```r
cat("1. The bank loan data shows a clear upward trend over the 70-month period\n")
```

```
## 1. The bank loan data shows a clear upward trend over the 70-month period
```

```r
cat("2. The data is non-stationary in its original form (confirmed by ADF test)\n")
```

```
## 2. The data is non-stationary in its original form (confirmed by ADF test)
```

```r
cat("3. First/second differencing successfully achieved stationarity\n")
```

```
## 3. First/second differencing successfully achieved stationarity
```

```r
cat("4. ARIMA(0,1,1) model provides a good fit to the data\n")
```

```
## 4. ARIMA(0,1,1) model provides a good fit to the data
```

```r
cat("5. Residual diagnostics confirm model adequacy (white noise residuals)\n")
```

```
## 5. Residual diagnostics confirm model adequacy (white noise residuals)
```

```r
cat("6. The model can be used for forecasting future bank loan values\n\n")
```

```
## 6. The model can be used for forecasting future bank loan values
```

```r
cat("PRACTICAL IMPLICATIONS:\n")
```

```
## PRACTICAL IMPLICATIONS:
```

```r
cat("- The fitted model captures the underlying pattern in bank loan data\n")
```

```
## - The fitted model captures the underlying pattern in bank loan data
```

```r
cat("- Can be used for short-term forecasting of loan amounts\n")
```

```
## - Can be used for short-term forecasting of loan amounts
```

```r
cat("- Helps in understanding the time series structure of financial data\n")
```

```
## - Helps in understanding the time series structure of financial data
```

```r
cat("- Provides basis for policy and decision making in banking sector\n\n")
```

```
## - Provides basis for policy and decision making in banking sector
```

```r
cat("LIMITATIONS:\n")
```

```
## LIMITATIONS:
```

```r
cat("- Model assumes linear relationships\n")
```

```
## - Model assumes linear relationships
```

```r
cat("- External economic factors not incorporated\n")
```

```
## - External economic factors not incorporated
```

```r
cat("- Model performance may degrade for long-term forecasts\n")
```

```
## - Model performance may degrade for long-term forecasts
```

```r
cat("- Requires periodic re-estimation as new data becomes available\n")
```

```
## - Requires periodic re-estimation as new data becomes available
```

```r
cat(rep("=", 70), "\n")
```

```
## = = =
```